

ASG Airlines: End-to-End Data Engineering Case Study

Introduction:

ASG Airlines project is an engineering solution built to process airline data & produce useful flight operation insights. As Azure was not a requirement the pipeline was built locally using Python and Pandas. The process includes data ingestion, data quality checking, data cleaning, data transformation, data validation, data aggregation and visualisation of data through Power BI.

Data Ingestion:

The source data was given in an excel workbook with four datasets: Flights, Bookings, Payments and Passengers. The initial datasets contained 1,020 flight records, 1,000 booking records, 1,000 payment records, and 1,039 passenger records.

Data quality checking:

The given data were checked for missing, duplicate, inconsistent values. The correctness & completeness of the overall data were analysed before data cleaning and transformation.

Data cleaning:

Duplicate flight records were removed to avoid double counting. Missing categorical values like airline and booking status were replaced with Unknown. Payment amounts were transformed to numeric values and invalid values were fixed while converting. Text fields such as airline, source, destination, and booking status were standardised by removing extra spaces and using consistent formats.

Data transformation:

One invalid flight record had a negative duration. Since a negative flight duration is not valid this record was removed from the duration analysis. The passenger and booking datasets contain sensitive information such as email addresses, phone numbers, Aadhaar IDs, passport numbers, and emergency contact details. Aadhaar ID and passport number were considered the most sensitive fields and they were protected using encryption.

Data validation:

Before using the data for final aggregation, the data is validated to ensure it meet the required conditions.

Data aggregation:

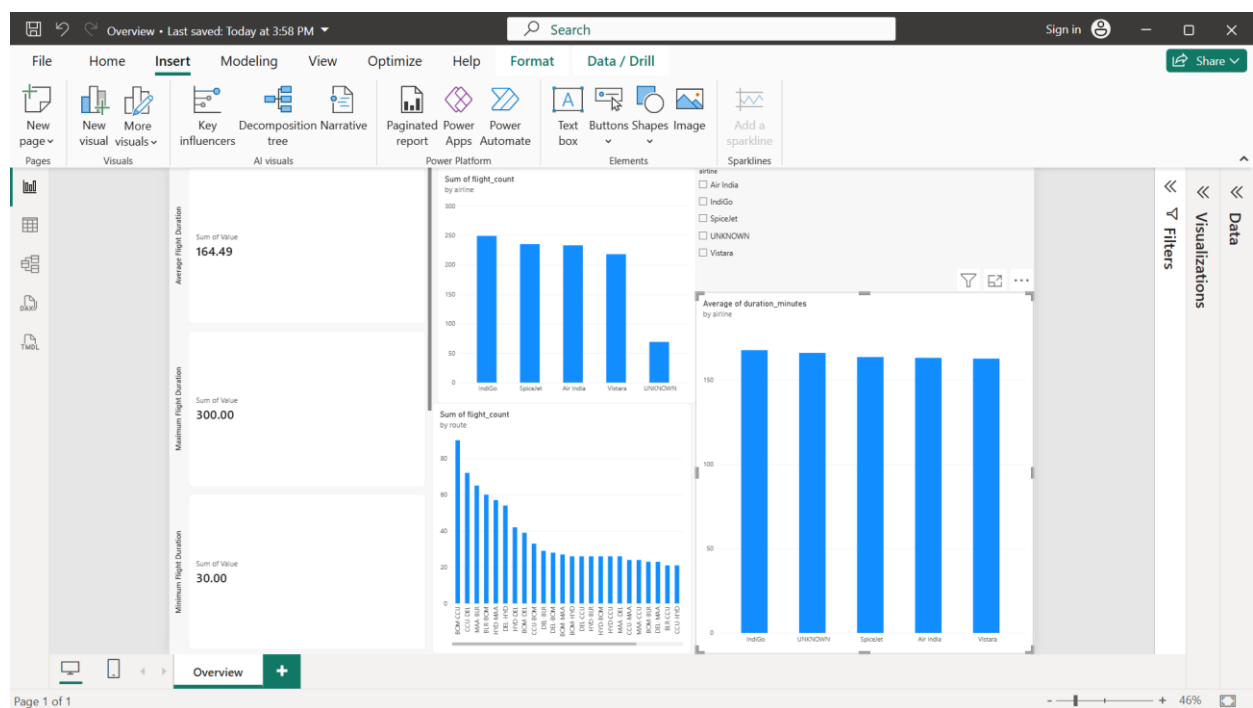
The cleaned data was aggregated into analysis-ready tables for Power BI.

- **KPI Summary** - total flights, average duration, total bookings, and payment-related KPIs.
- **Route Traffic** - number of flights for each source-destination route.
- **Airline Trends** - number of flights operated by each airline.
- **Duration Analysis** - flight, airline, route, and calculated duration information.
- **Anomaly Analysis** - records identified as flight timing anomalies.

These tables were exported as CSV files and used as the data source for Power BI.

Data visualisation:

Power BI Desktop was used to create an interactive flight operations dashboard. The dashboard contains KPI cards showing important summary values, charts for airline and route performance, and a chart comparing average flight duration across airlines.



Data flow architecture:

